POT-C

The smart trash bin that transforms your organic waste into natural compost, without any chemicals.

Process of the Change

Who We Are

We are POT-C, and we created a project that seeks to make a positive environmental change through technological solutions accessible to Salvadoran families. Our solution is a smart trash bin whose main function is to produce natural/organic compost through automated processes.

Vision

To be the driving force that turns every urban kitchen into a sustainability hub, where composting becomes a natural daily habit that inspires and builds a community of change, making El Salvador greener, more conscious, and resilient.

Mission

Our project promotes the proper use of organic waste, focusing on the composting method for adults interested in environmental care through innovative tools.

What is Composting?

Biological process in which, under suitable conditions, microorganisms break down organic waste and then transform it into natural fertilizer for your plants.

Benefits of Organic Composting

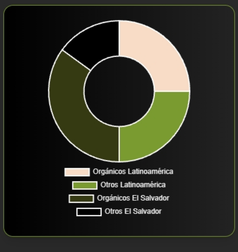
- **Keeps soil healthy and nourishes plants:** Unlike chemical fertilizers that only feed the plants but degrade the soil, organic compost does both.
- **Savings:** Reducing reliance on chemical fertilizers in the long term can lead to significant savings.
- **Giving new life to your waste:** Finally, you can give a new use to those fruit or vegetable peels that you normally throw away.

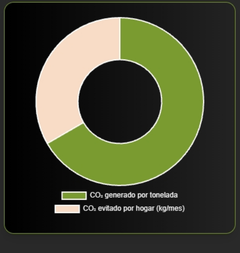
Problem Data

We present a visual and detailed summary of the organic waste problem, its environmental impact, and our proposed solution to promote composting in Salvadoran households.

The Problem

- More than 50% of urban waste in Latin America is organic, but only 2% is properly utilized (Carvalho & Cirión, 2022).
- Each ton of mismanaged organic waste generates 0.5 tons of CO_2 equivalent.
- In Salvadoran schools and homes, over 70% of waste is organic, yet it is rarely separated or utilized.





Environmental Impact

- Reducing organic waste means less pollution and fewer greenhouse gases.
- With just one household composting, up to 250 kg of CO_2 can be avoided per month.
- Compost improves soil health, reducing reliance on chemical fertilizers.

Our Solution

- Compact bin that doesn't take up much space.
- Automatically shreds and controls aeration, oxygen, and temperature.
- 5-phase controlled process, odor-free and hassle-free.
- Produces high-quality compost in approximately 15 days.

Want to Know More?

Discover more information about our project and how you can help reduce organic waste in your home.

See More

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Smart trash bin project that transforms organic waste into natural compost. Inspiring homes and schools to care for the environment.

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LEARN MORE ABOUT OUR PROJECT

Don't know how to make good use of your organic waste?

Our solution: **Klani.**

Dedicated to the automated creation of **100% natural** compost, taking advantage of the properties of organic waste to achieve results in up to **15 days**.

How does it work?

Turn Your Trash into Life

Imagine a trash bin that not only stores your waste... **but transforms it into a high-quality natural fertilizer!**

Our smart bin combines oxygen, pH, gas, and temperature sensors, along with aeration systems to create **fast compost, odor-free, with low energy consumption, and in the simplest way.**

Designed for Salvadoran families, this project not only saves on fertilizers but also transforms your lifestyle into one that is cleaner, more natural, and more responsible with the planet. This innovative system turns what was once waste into a valuable resource for your home.

Easy, affordable, and eco-friendly.
Because taking care of the planet can also start at home.

Features

Fast and Efficient Process
Thanks to its technology, the bin accelerates the decomposition of organic waste, reducing traditional composting times and delivering visible results in less time.

Easy to Use
Its intuitive system allows any family member to operate it without complications. No prior knowledge of composting is required.

Weed and Odor Resistant Design
Thanks to aeration and temperature control, waste is transformed without producing unpleasant odors, making its use comfortable and pleasant inside the home.

← →

3D Model

Below we provide you with a look at the 3D model of our trash bin: Klani.

Show 3D Model

Composting Phases

The Klani bin features an automated 5-stage process, where each one performs a different action simultaneously with the goal of obtaining organic compost in approximately 15 days.

- **Bin (Peels):** The beginning of the process: your organic waste, such as peels and kitchen scraps, is automatically shredded by the bin.
- **Mesophilic Phase:** During the first days, microorganisms begin breaking down the materials at moderate temperatures, accelerating the start of composting.
- **Thermophilic Phase:** The temperature naturally rises, eliminating unwanted bacteria and breaking down tougher waste, ensuring clean and safe compost.
- **Compost Resting:** After the intense heat, the compost rests and matures, stabilizing its nutrients and achieving the ideal texture and consistency.
- **Final Compost Product:** Finally, you obtain a natural, nutrient-rich fertilizer ready to nourish your plants. What was once waste is transformed into life, savings, and sustainability for your home.

Trash Bin Benefits

1
Fast and Automated
The trash bin accelerates the decomposition of organic waste automatically, reducing traditional times.

2
Easy to Use
Its intuitive system allows any family member to operate it without complications.

3
Fits Small Spaces
Its compact size allows it to be easily placed in any corner of the house, adapting to apartments or small kitchens.

Would You Like Some Recommendations?

Learn how to use our trash bin: Klani, more effectively with the following recommendations for proper use.

Tips

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RECOMMENDATIONS

Do you want to learn how to make the most of our smart trash bin?

Here we present some recommendations and tips to get the best use out of our bin, allowing you to produce 100% organic compost in approximately 15 days.

What to Use?

Organic Waste Options to Use

Essentially, seeds, vegetables, and fruits are good ingredients for quality compost. However, it is important to know that each food item may have a more specific result.

Recommended to Use

- Fruit and vegetable scraps.
- Crushed eggshells.
- Coffee grounds and filters.
- Dry leaves and cut grass.
- Cardboard without ink and plain paper.

Use in Moderation

- Citrus peels in excess.
- Bread, rice, or pasta (small amounts).
- Banana peels (cut).
- Lightly cooked food scraps.
- Old bread or plain cookies.

Do Not Use

- Meat and bones.
- Oils and fats.
- Dairy products.
- Pet waste.
- Plastics, metals, glass.

What to Avoid?

- **Dairy:** Has a very high pH which causes loss of essential nutrients like nitrogen, and makes it harder for plants to absorb nutrients.
- **Citrus:** Makes the compost pH too acidic and low, slowing down the composting process.

Frequently Asked Questions

⏱️ **How long does the composting process take?** It takes around 15 days, with each phase lasting 5 days.

⚠️ **What happens if I put something I shouldn't in the composter?** It depends on what was added, but most commonly it affects the pH, slows down the process, creates bad odors, or even attracts insects.

🛠️ **Do I need to buy anything else for the robot to work?** No. Our compost robot is designed to be a complete and self-sufficient solution. You just need to add your organic waste and wait.

🌿 **What do I do with the compost once it's ready?** Once ready, you can apply it directly to plants. If there's any left, store it in a bag in a dry place to prevent moisture from ruining it.

🌱 **How do I know my compost is perfect?** Poorly made compost can still have food chunks or be damp, whereas good compost looks like soil, with no large food residues and completely dry.

App

We also have our own mobile application designed to make using our smart bin easier. In the app, you can find detailed information such as temperature, pH, oxygen, humidity, and gas levels. It allows better control over the bin, showing how much time is left for the processes to finish, along with extra information.

⚠️ Currently, we only have a prototype of the app in Figma. You can check it out through the button below ⚠️

Open in Figma

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